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Digital Electronics & VLSI
Codec Technologies India
Internship 2026

UART Protocol Engine in VHDL-2008

16x oversampling TX/RX FSMs, parity, 2 stop bits, RX FIFO with overflow, baud-mismatch tolerance

AT A GLANCE

68 / 68 pass

GHDL checks

46 at 5 rates

Frames simulated

9600 / 115200 / 921600

Loopback rates

4.5 % ok, 13 % errors

Mismatch tolerance

378 / 7680

iCE40 logic cells (nextpnr)

134.95 MHz

iCE40 Fmax (nextpnr)

9.15 ns

icetime longest path

107 / 101 / 2

Xilinx LUT / FF / RAM32M

VERDICT

SIM PASS

P&R OK

TIMING MET

GHDL PLUGIN USED

DECLARED SUBSTITUTION

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1. Objective and brief mapping

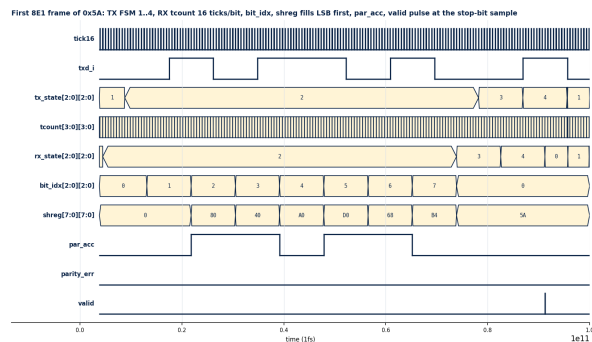
The brief asked for a VHDL-2008 UART with separate TX and RX FSMs, a parameterised 16x oversampling baud generator, 8N1 plus optional even/odd parity and 1 or 2 stop bits, framing and parity error flags, an 8-deep RX FIFO with an overflow flag and a loopback top; a testbench covering loopback at 9600, 115200 and 921600 baud, an injected parity error, a framing error from a line held low, FIFO overflow and a baud mismatch (RX 115200 with TX 110000 decoding and TX 100000 failing), all self-checking; waveform PNGs of a frame with the parity error highlighted; synthesis as in project 03; and a pyserial PC terminal, a Basys3 USB-UART .xdc and a ModelSim .do, declared not run.

All of it is implemented. Every number here comes from GHDL 7.0.0-dev, yosys 0.68 with the ghdl plugin, nextpnr-ice40 and icetime in the build container. pyserial, Vivado and ModelSim were not available; section 6 states exactly what was substituted.

2. Architecture

```
clk -> baud_gen(BAUD_TX): acc += 16*BAUD, tick16 on wrap -> uart_tx IDLE>START>DATA x8>[PARITY]
clk -> baud_gen(BAUD_RX)                               -> uart_rx 2-FF sync, start edge, ver
                                                         data[7:0], valid, parity_err, frame
                                                         -> sync_fifo 8 x {ferr, perr, da
inj_perr: inverts TX parity sense (
```

Block	Design points
baud_gen	Phase accumulator instead of a divider: $12e6 / (16 \times 115200) = 6.51$ would be 7.4 % off as an integer divider; the accumulator's average tick rate is exactly $16 \times \text{BAUD}$ with at most one clock of jitter. Asserts $16 \times \text{BAUD} < \text{CLK_HZ}$ at elaboration (this is what forced a 48 MHz clock for 921600 baud).
uart_tx	Configuration (parity, stop bits) is latched with the byte so a change mid-frame cannot corrupt it; parity is computed at load; a brk input drives the line low for framing-error tests; done pulses at the end of the last stop bit.
uart_rx	Start edge in IDLE, re-checked at the centre of the start bit to reject glitches; each bit sampled 16 ticks after the previous sample point; parity compared against the configured sense; stop bit(s) must be 1 or frame_err is set; a break gives 0x00 + frame_err.
sync_fifo	8 x 10-bit first-word-fall-through FIFO storing the two flags with each byte, so rx_perr / rx_ferr always describe the word at the head; a write when full is dropped and sets the sticky overflow flag; simultaneous read + write when full succeeds.



Receiver internals for the first 0x5A frame: tcount counts 16 ticks per bit, bit_idx steps through D0..D7, shreg fills LSB first (5A appears after the eighth sample), par_acc accumulates the parity, valid pulses at the stop-bit sample.

3. Verification

tb_uart.vhd instantiates uart_top five times on one 48 MHz clock (12 MHz oscillator x4 PLL): 9600/9600, 115200/115200, 921600/921600, TX 110000 / RX 115200 and TX 100000 / RX 115200. Every expectation is an assert with severity failure, so a failing check stops GHDL with a non-zero exit code and run_all.sh fails. Because external names crash at run time in this GHDL build, the bench observes the design only through its ports and uses the inj_perr and tx_brk hooks to create errors. Result: 46 frames, 68 checks, PASS.

Test	Instance	Checks	Result
T1 loopback 8N1	9600, 115200, 921600	0x55 0xA3 0x00 0xFF received in order, no flags, FIFO empty afterwards	pass
T2 parity	115200	8E1 and 801 clean; inj_perr on 0x5A -> rx_perr = 1 with data intact and no frame error; flag not sticky on the next byte	pass
T3 framing	115200	tx_brk for 12 bit periods -> rx_ferr = 1, data 0x00; after release the receiver resynchronises and 0xA5 arrives clean	pass
T4 two stop bits	115200	8N2 loopback clean	pass
T5 FIFO overflow	921600	10 frames unread: count 8, full, overflow sticky; first 8 in order; 9 and 10 dropped; clr_ovf clears the flag	pass
T6 mismatch 4.5 %	TX 110000 / RX 115200	4 bytes decode cleanly	pass
T6 mismatch 13 %	TX 100000 / RX 115200	5 frames seen (0xAD, 0xB5, 0xD5, 0xAD with ferr, 0xF5 clean), none equal to the sent 0x55	pass

KEY TAKEAWAY A phase-accumulator baud generator and a re-synchronising 16x receiver give a UART that decodes a 4.5 % rate mismatch, flags parity/framing errors per word in the FIFO, and closes 135 MHz on an HX8K.

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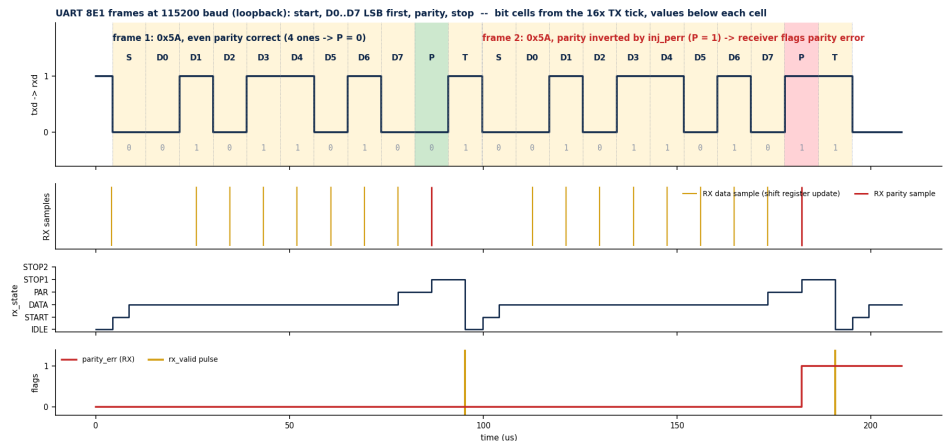
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Two 8E1 frames of 0x5A at 115200 baud from the GHDL VCD. Bit cells come from the transmitter's 16x tick; the first frame carries the correct even parity (green cell), the second has its parity inverted by inj_perr (red cell). The receiver samples each bit at its centre (gold), samples parity on the PAR to STOP1 transition (red), and parity_err rises exactly with the second rx_valid pulse.

4. Synthesis and implementation results

The two baud accumulators are 24-bit adders with a compare, which is where most of the carry chains and about a third of the flip-flops go; the FSMs themselves are small. On Xilinx the FIFO memory maps to two RAM32M distributed-RAM primitives, on the iCE40 yosys keeps the 8 x 10-bit array in logic (no BRAM used). Timing is comfortable everywhere: at 12 MHz the slack is over 70 ns, and the 48 MHz PLL clock needed for 921600 baud is still 2.8x below the placed Fmax.

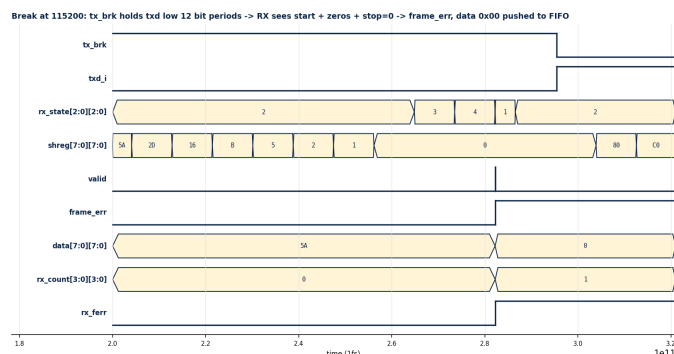
Target / tool	Resources	Timing
iCE40-HX8K, yosys -m ghdl synth_ice40 (12 MHz, 115200)	234 SB_LUT4, 181 FF (2 DFF, 106 DFFE, 42 DFFESR, 31 DFFSR), 71 SB_CARRY	-
iCE40-HX8K CT256, nextpnr-ice40, real .pcf (FTDI pins B12/B10)	378 / 7680 logic cells (4 %), 47 / 206 IO	Fmax 134.95 MHz, 12 MHz PASS
iCE40-HX8K, icetime -mtr	4 logic levels	9.15 ns (109.24 MHz)
Xilinx xc7 (Basys3 class), yosys synth_xilinx	107 LUT, 101 FF, 25 CARRY4, 2 RAM32M for the FIFO	not placed (no Vivado)

Frame budget for the configurations exercised, with the clocks per 16x tick that the accumulator has to realise on the three clocks used in this project (12 MHz HX8K oscillator, 48 MHz testbench/PLL, 100 MHz Basys3); non-integer values are exactly where a plain divider would introduce the rate error the accumulator avoids:

baud	format	bits/frame	frame time	throughput	clk/tick @12 MHz	@48 MHz	@100 MHz
9600	8N1	10	1041.67 us	960 B/s	78.12	312.50	651.04
115200	8N1	10	86.81 us	11520 B/s	6.51	26.04	54.25
115200	8E1	11	95.49 us	10473 B/s	6.51	26.04	54.25
115200	8N2	11	95.49 us	10473 B/s	6.51	26.04	54.25
921600	8N1	10	10.85 us	92160 B/s	0.81	3.26	6.78

5. Waveforms and mismatch behaviour

The mismatch tolerance follows from the sampling geometry: the receiver re-aligns on every start edge and samples the stop bit 9.5 receiver bit-periods later, so the accumulated error is 9.5 x (rate error). At 110000 vs 115200 (4.5 %) the stop bit is sampled 0.43 bit-periods early, still inside the stop cell, and all data bits are also sampled inside their cells: four clean bytes. At 100000 vs 115200 (13 %) the stop sample lands 1.2 bit-periods early, inside D7, and the data samples walk backwards into earlier bits: the receiver reported 0xAD, 0xB5, 0xD5, 0xAD with frame errors and one clean 0xF5, none of them the transmitted 0x55, which is exactly the failure the brief asked to demonstrate.



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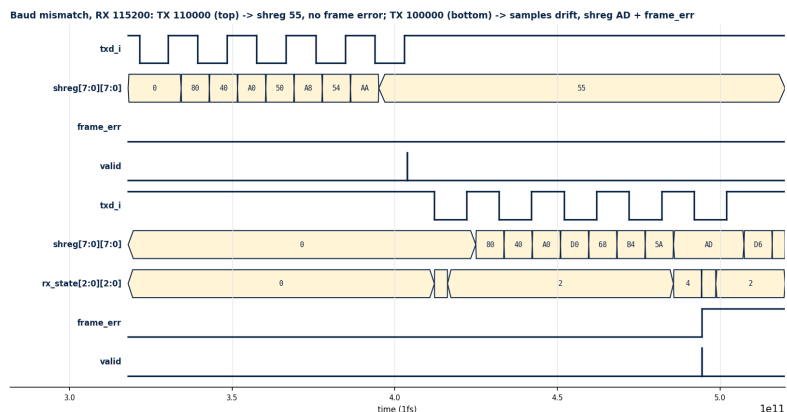
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Break test: tx_brk holds txd low for 12 bit periods; the receiver takes it as a start bit, shifts in zeros, finds the stop bit low and pushes 0x00 with frame_err into the FIFO (rx_count 0 -> 1, rx_ferr = 1 at the head).

Sample positions in transmitter bit periods for the two mismatch cases (receiver samples bit k at (k + 1.5) receiver periods after the start edge; the transmitter's cell for bit k spans k + 1 to k + 2):

bit	TX cell (TX bit periods from start edge)	RX sample, TX 110000 (4.5 %)	RX sample, TX 100000 (13 %)
D0	1.0 - 2.0	1.43	1.30
D1	2.0 - 3.0	2.39	2.17
D2	3.0 - 4.0	3.34	3.04
D3	4.0 - 5.0	4.30	3.91
D4	5.0 - 6.0	5.25	4.77
D5	6.0 - 7.0	6.21	5.64
D6	7.0 - 8.0	7.16	6.51
D7	8.0 - 9.0	8.12	7.38
stop	9.0 - 10.0	9.07	8.25



Both mismatch instances receiving 0x55 (alternating bits) in the GHDL VCD. Top, TX 110000: the shift register ends at 0x55 and valid fires without frame_err (stop sampled at 9.07 TX bit periods, just inside the 9.0-10.0 stop cell). Bottom, TX 100000: the samples walk into earlier bits, shreg ends at 0xAD and the stop sample at 8.25 falls inside the D7 cell (a 0 for 0x55), so frame_err rises with valid, exactly as the table predicts.

5b. FIFO overflow test (T5) as observed

At 921600 baud a frame lasts 10.85 us, so ten back-to-back frames arrive in 109 us while the consumer reads nothing; the FIFO must keep the first eight, refuse the rest and remember that it did.

step	observed
frames sent (921600, none read)	0x40 0x41 0x42 0x43 0x44 0x45 0x46 0x47 0x48 0x49 (10)
after the burst	rx_count = 8, rx_full = 1, rx_overflow = 1 (sticky)
8 reads	0x40 .. 0x47 in order, each with perr = 0, ferr = 0
after the reads	rx_empty = 1, frames 0x48 and 0x49 lost, rx_overflow still 1
clr_ovf pulse	rx_overflow = 0

6. Declared substitutions and target-tool artefacts

- pyserial / PC: not installed and no board attached. host/pc_terminal.py imports pyserial lazily inside open_port(), so --help and tests/test_pc_terminal.py (3 unittest cases on the pure functions) run here; the serial paths --loopback, --send and --terminal were not exercised against hardware.
- Vivado / Basys3: not installed. vivado/basys3.xdc maps rxd/txd to the USB-UART bridge (B18 RsRx, A18 RsTx), configuration to switches, tx_data to SW8-15, rx_data and flags to LEDs; vivado/create_project.tcl sets CLK_HZ = 100 MHz and builds the bitstream. The Xilinx LUT/FF/RAM figures are from yosys synth_xilinx, not Vivado.
- ModelSim / Questa: not installed. modelsim/run.do compiles the sources with vcom -2008 and runs tb_uart; all simulation results here are from GHDL.
- No board: iCE40 figures are from nextpnr/icetime on the HX8K device database; 921600 baud on the 12 MHz breakout needs a PLL to 48 MHz, which the .pcf and top do not include (the testbench uses a 48 MHz clock for that reason).

7. Learnings and next steps

Storing the error flags in the FIFO word rather than in a status register was the most useful decision: with several frames queued, a global parity flag would be meaningless, and the testbench checks that the flag belongs to the right word (clean, flagged, clean). The break test also taught me to think about what the receiver does after a framing error: it returns to IDLE, sees the line still low and starts another frame, so the bench must wait for that second frame before expecting clean data. Next steps: a TX FIFO with a CTS/RTS flow-control pair, a programmable baud register so one bitstream serves all rates, and a Vivado run on Basys3 with the pc_terminal.py loopback test at 921600 baud.

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